Data Structures and Algorithms Project
Gomoku (Five in a Row) Game with AI On a 9 x 9 board

DUE DATE: Friday 4th APRIL 2025

This Project is worth 15%

Project Group Size: 4 to 5 persons

NOTE: Students in the Friday lab session MUST choose members from their session (their program) only.

This project has two distinct parts:

The submitted project document: 7.5%

Presentation/ student interview: 7.5%

All student group interviews will be done in the April 7th to 11th week during the Lab session

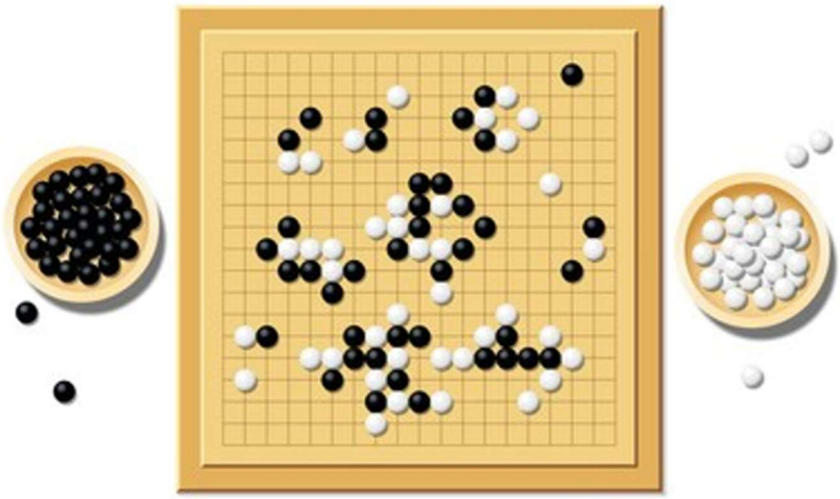
1 Overview

1.1 Project Steps

You must create a fully working intelligent command-line Gomoku (Five in a Row) game in Java using only basic arrays (e.g., char[][] or int[][]). Higher-level data structures beyond simple arrays will result in a maximum achievable mark of 50% for this assignment. The game must allow the user to select the mode of the game. The two modes are:

- 2 player (2 human players)
- 1 player (Human vs AI)

Gomoku (Five in a Row) Game:
Gomoku is a classic two-player strategy board game often played on a 15×15 grid (though smaller or larger boards are also common). Each player selects a color—traditionally black or white stones—and takes turns placing a single stone on an empty intersection. The goal is to be the first to arrange five stones in a continuous line—horizontally, vertically, or diagonally. Once a player achieves five in a row, the game ends immediately and that player is declared the winner. This project will use a 9 x 9 grid.



A detailed overview of the Gomoku (Five in a Row) game can be found here:
https://www.youtube.com/watch?v=nenFRLFY7yc

This logic for the AI must be implemented using the strategy outlined below.

Intelligent AI:
This **must** be implemented using the **Minimax algorithm**. The computer must examine the game state at each point and deduce the optimal (best) play that can be made.

A detailed overview of the Minimax algorithm can be found here:
https://www.youtube.com/watch?v=zp3VMe0Jpf8

The Game should operate as follows:

1) The user should be asked if it is a 2 player OR 1 player game (versus AI)
2) If the user selects **1 player (versus minimax AI)**
    a. The Human player should be asked their name
    b. The player must be prompted to choose their symbol ('B' or 'W') (B for black, W for white)
    c. The black player should go first (computer or human player).
    d. The present board state will be displayed and:
        1. If the computer is to play, it will now show the board updated with the computer's symbol.
        2. If it is the player's turn, the player will be prompted for a column location to play.
    e. The board will then be checked for a winning state or draw and if it is not in a winning state or draw(tie), go to step **d** otherwise display the winner or declare a draw (tie).

3) If the user selects **2 player (2 Human players)**
    f. Each human player must be prompted to choose their name and symbol ('B' or 'W') (B for black, W for white)
    g. An empty board will then be displayed.
    h. The player next to move will be prompted for a column location to play.
    i. The present board state will be displayed showing their updated play.
    j. The board will then be checked for a winning state or draw and if it is not in a winning state or draw, go to step **h** otherwise display the winner or declare a draw.

At the end of a game, the result should be displayed.

YOU MUST ONLY USE BASIC ARRAYS TO MODEL THE BOARD STATE – if this is not done, the maximum mark that can be attained is 5/10.

NOTE:

The non-AI part (2 human players) is worth 70% of the marks

The recursive minimax version (1 human vs AI) is worth 30% of the marks

This assignment loses 30% each day (based on the date) after the 4th April.

This means if it is submitted 12:01 AM (1 minute late) on the 5th, the maximum mark that can be got is 7/10.

This means if it is submitted 12:01 AM on the 6th, the maximum mark that can be got is 4/10.

It is therefore better to submit a non-working AI version on time than an AI version late.

1.2 Submission Format

This is a group assignment, and one report and code will be submitted per group.

Your group must submit two Items:

1) The completed submission document (Given with this project) with ALL the group member's names inserted at the top and code pasted neatly into it.

2) The completed Java source files (zipped) with all student's names and numbers commented at the top of the code[not the whole project, just all the java classes you created]. **To be clear, do NOT upload a whole solution just the classes**.

Please note your code must be properly documented.